Online supplement section 1

Functional groups	Sources and References
1. Phytoplankton (spp.)	
B	From Chl-a via satellite (http://emis.jrc.ec.europa.eu/#.). Conversion factors from Jorgensen et al. (1991) and Dalsgaard and Pauly (1997)
P/B	Primary production from via satellite data (http://emis.jrc.ec.europa.eu/#). Conversion factor from Dalsgaard and Pauly (1997)
2. Benthic primary producer (spp.)	
B	Estimated using a EE = 0.5 (1990-1994) and 0.7 (2008-2012). We assumed a higher EE in 2008-2012 due to the impact of alien herbivores
PB	Corrales et al. (2015). Data corrected following Opitz (1996)
3. Micro and Mesozooplankton (spp.)	
B	Terbiyik Kurt and Polat (2013)
PB	Siokou-Frangou et al. (2002) and Zervoudaki et al. (2011)
QB	Estimated using a P/Q = 0.33 (Hansen et al., 2000)
Diet	Calbet et al. (2002) and Zervoudaki et al. (2007)
4. Macrozooplankton (euphasids, pteropods and fish larvae and eggs)	
B	Estimated using a EE = 0.95
PB	Labat and Cuzin-Roudy (1996). Data corrected following Opitz (1996)
QB	Baamstedt and Karlson (1998). Data corrected following Opitz (1996)
Diet	Baamstedt and Karlson (1998)
5. Gelatinous plankton (cubozoa, hydrozoa, scyphozoa, tunicata)	
B	Estimated using a EE = 0.3
PB	Malej (1989). Data corrected following Opitz (1996)
QB	Malej (1989). Data corrected following Opitz (1996)
Diet	Graham and Kroutil (2001) and Orek (2000)
6. Polychaetes (nematods and annelids)	
B	Mutlu et al. (2010)

PB	Serrano (2012) and Sardá et al. (2000). Data corrected following Opitz (1996)
QB	From the empirical equation of Cammen (1979)
Diet	Fauchald and Jumars (1979)

7. Suprabenthos (mysids, isopods, amphipods, cumaceans and copepods)

B	Estimated using a EE = 0.8
PB	Corbera et al. (2013). Data corrected following Opitz (1996)
QB	Cartes and Maynou (2001). Data corrected following Opitz (1996)
Diet	Cartes et al. (2001)

8. Native shrimps (*Aristeus antennatus, Aristaemorpha foliacea, Parapennaeus longirostris, Plesionika martia* and *Squilla mantis*)

B	Bottom trawl surveys (Edelist et al., 2011; Edelist et al., 2013a)
PB	Tsagarakis et al. (2010). Data corrected following Opitz (1996)
QB	From the empirical equation of Pauly et al. (1990). W∞ from Abelló and Sardá (1989), Levi et al. (1995) Papaconstantinou and Kapiris (2001), Maiorano et al. (2002) and Papaconstantinou and Kapiris (2003),
Diet	Froglia and Giannini (1989), Cartes (1993), Chartosia et al. (2005), Nouar et al. (2011) and Cartes et al. (2014)

9. Alien shrimps (*Marsupennaeus japonicus, Metapenaeopsis aegyptia, Metapenaeus monoceros, Penaeus semisulcatus* and *Trachysalambria curvirostris*)

B	Bottom trawl surveys (Edelist et al., 2011; Edelist et al., 2013a)
PB	Tsagarakis et al. (2010). Data corrected following Opitz (1996)
QB	From the empirical equation of Pauly et al. (1990). W∞ from Sukumaran et al. (1993), Ye (1998), Fatima (2001) and Hossain and Ohtomi (2010)
Diet	Same than the native group

10. Native crabs (*Alpheus spp., Callinectes sapidus, Calappa granulata, Dardanus arrosor, Inachus dorsettensis, Latreilla elegans, Medorippe lanata* and *Scyllarides lateus*)

B	Bottom trawl surveys (Edelist et al., 2011; Edelist et al., 2013a)
PB	Tsagarakis et al. (2010). Data corrected following Opitz (1996)
QB	From the empirical equation of Pauly et al. (1990). W∞ from Bianchini et al. (2001) and Sumer et al. (2013)
Diet	Tsagarakis et al. (2010) and Corrales et al. (2015)

11. Alien crabs (*Charybdis longicollis, Erugosquilla massavensis, Ixa monody, Myra subgranulata* and *Portunus pelagicus*)

B	Bottom trawl surveys (Edelist et al., 2011; Edelist et al., 2013a)

PB	Tsagarakis et al. (2010). Data corrected following Opitz (1996)
QB	From the empirical equation of Pauly et al. (1990). W∞ from Gokoglu et al. (2008) and Sangun et al. (2009)
Diet	Same than the native group

12. Benthic invertebrates (echinoderms, bivalves, gastropods, cnidarians and sponges)

B	Estimated using a EE = 0.8
PB	Moodley et al. (1998). Data corrected following Opitz (1996)
QB	Brey (2001)
Diet	Millar (1971), Perron and Turner (1978), Lalli and Gilmer (1989), Coulon and Jangoux (1993), Juan et al. (2007) and Baeta and Ramón (2013)

13. Benthic cephalopods (*Octopus vulgaris, Sepia elegans, Sepia officinalis, Sepia orbignyana* and *Sepietta oweniana*)

B	Bottom trawl surveys (Edelist et al., 2011; Edelist et al., 2013a)
PB	Corrales et al. (2015). Data corrected following Opitz (1996)
QB	Iglesias et al. (1996) and Quintela and Andrade (2002)
Diet	Quetglas et al. (1998), Alves et al. (2006) and Neves et al. (2009)

14. Benthopelagic cephalopods (*Illex coindetii* and *Loligo vulgaris*)

B	Bottom trawl surveys (Edelist et al., 2011; Edelist et al., 2013a)
PB	Coll et al. (2013)
QB	Coll et al. (2013)
Diet	Pierce et al. (1994) and Rosas-Luis et al. (2013)

15. Mullets (*Mullus barbatus* and *Mullus surmuletus*)

B	Bottom trawl surveys (Edelist et al., 2011; Edelist et al., 2013a)
PB	Z=F+M; M= empirical equation from Pauly (1980). L∞ and K from Mehanna (2009c) and Mehanna (2009a)
QB	From the empirical equation of Pauly et al. (1990). W∞ from Edelist et al. (2013c)
Diet	Golani (1994)

16. Goatfishes (*Upeneus moluccensis* and *Upeneus pori*)

B	Bottom trawl surveys (Edelist et al., 2011; Edelist et al., 2013a)
PB	Z=F+M; M= empirical equation from Pauly (1980). L∞ and K from Ismen (2005) and Işmen (2006)

QB	From the empirical equation of Pauly et al. (1990). W∞ from Edelist (2014)
Diet	Golani (1994)

17. Hake (*Merluccius merluccius*)

B	Bottom trawl surveys (Edelist et al., 2011; Edelist et al., 2013a)
PB	Z=F+M; M= empirical equation from Pauly (1980). L∞ and K from Mehanna (2009b)
QB	From the empirical equation of Pauly et al. (1990). W∞ from Mehanna (2009b)
Diet	Philips (2012)

18. Flatfishes (*Arnoglossus spp., Bothus podas, Buglossidium luteum, Citharus linguatula, Cynoglossus sinusarabici, Microchirus spp., Solea solea* and *Symphurus nigrescens*)

B	Bottom trawl surveys (Edelist et al., 2011; Edelist et al., 2013a)
PB	Z=F+M; M= empirical equation from Pauly (1980). L∞ and K from Schintu et al. (1994), Ozutok and Avsar (2002) Turkmen (2003) and Türker Çakir et al. (2005)
QB	From the empirical equation of Pauly et al. (1990). W∞ from Abdallah (2002), Turkmen (2003), Sangun et al. (2007) and Edelist (2014)
Diet	Bayhan et al. (2009), Karachle and Stergiou (2011) and Abid et al. (2013)

19. Rocky fishes (*Apogon imberbis, Chromis chromis, Blennius ocellaris, Diplodus annularis, D. cervinus, D. puntazzo, D. sargus, D. vulgaris* and *Gobius spp*.)

B	Bottom trawl surveys (Edelist et al., 2011; Edelist et al., 2013a)
PB	Z=F+M; M= empirical equation from Pauly (1980). L∞ and K from Garnaud (1962), Dulčić and Kraljević (1995), (Bradai et al., 1998), and (Pajuelo et al., 2003b) and Apostolidis and Stergiou (2014)
QB	From the empirical equation of Pauly et al. (1990). W∞ from Karakulak et al. (2006), Sangun et al. (2007), Edelist et al. (2013c) and Edelist (2014)
Diet	Pölzer and Patzner (1998) Labropoulou and Papadopoulou-Smith (1999), Derbal et al. (2007), Benchalel et al. (2010), Lipsky (2012) and Chaouch et al. (2013)

20. Small native demersal fishes (*Ariosoma balearicum, Balistes capriscus, Callionymus risso, Cepola macrophthalma, Chelidonichthys cuculus, Chelidonichthys lucerna, Dactylopterus volitans, Deltentosteus quadrimaculatus, Echelus myrus, Echeneis naucrates, Eutrigla gurnardus, Lepidotrigla cavillone, L. dieuzeidei, Lithognathus mormyrus, Liza aurata, L. ramada, Mugil cephalus, Oblada melanura, Pagellus acarne, P. bogaraveo, P. erythrinus, Pagrus caeruleostictus, P. pagrus, Phycis blennoides, P. physis, Pomadasys incisus, Sciaena umbra, Scorpaena notata, S. porcus, S. scrofa, Serranus cabrilla, S. hepatus, Sparus aurata, Synodus saurus, Trachinus araneus, Trigla lyra, Trigloporus lastoviza, Uranoscopus scaber* and *Zeus faber*)

B	Bottom trawl surveys (Edelist et al., 2011; Edelist et al., 2013a)
PB	Z=F+M; M= empirical equation from Pauly (1980). L∞ and K from Rafail et al. (1968), Papaconstantinou (1982), Cavallaro et al. (1985), Bradai and Bouain (1988), Erzini (1991), Papaconstantinou et al. (1993), Koutrakis and Sinis (1994), Faltas and Abdallah (1997), Tosunoglu et al. (1997), Stergiou

	et al. (1998), Chakroun-Marzouk (1999), Chakroun-Marzouk and Ktari (2001), Kaya et al. (2001), Pajuelo et al. (2003a), Türkmen and Akyurt (2003), Ismen et al. (2004), Mesa et al. (2005), Mehanna (2007), Bilgin and Çelik (2009), El -Hawwet et al. (2009), Ilhan et al. (2010), Ismen et al. (2013) and Soykan et al. (2013)
QB	From the empirical equation of Pauly et al. (1990). W∞ from Papaconstantinou et al. (1993), Petrakis and Stergiou (1995), Stergiou and Moutopoulos (2001), Morato et al. (2001), Abdallah (2002), Cicek et al. (2006), Karakulak et al. (2006), Mehanna (2007), Sangun et al. (2007), El -Hawwet et al. (2009), Edelist et al. (2013c) and Edelist (2014)
Diet	Caragitsou and Papaconstantinou (1988), Papaconstantinou and Caragitsou (1989), Papaconstantinou et al. (1989), Bradai and Bouain (1988), Caragitsou and Papaconstantinou (1994), Labropoulou and Eleftheriou (1997), Labropoulou and Machias (1998), Terrats et al. (2000), Morte et al. (2001), Pallaoro et al. (2004), Chaoui et al. (2005), Kallianiotis et al. (2005), Boudaya et al. (2007), Derbal and Kara (2007), Fehri and Gharbi (2008), Ben Hadj Hamida et al. (2009), Bilecenoglu (2009), Fehri-Bedoui et al. (2009), Esposito et al. (2009), Montanini et al. (2010), Stagioni et al. (2012) and Ismen et al. (2013)

21. Large native demersal fishes (*Argyrosomus regius, Epinephelus aeneus, E. costae, E. marginatus, Hyporthodus haifensis, Dentex dentex, D. gibbosus, D. macropthalmus* and *D. maroccanus*)

B	Bottom trawl surveys (Edelist et al., 2011; Edelist et al., 2013a)
PB	Z=F+M; M= empirical equation from Pauly (1980). L∞ and K from Rafail and Jukic (1971), Rafail (1972), Ezzat et al. (1981), Mennes (1985), Kara and Derbal (1995) and Chemmam-Abdelkader et al. (2004)
QB	From the empirical equation of Pauly et al. (1990). W∞ from Rafail (1972), Ceyhan et al. (2009), Torres et al. (2012), Edelist et al. (2013c) and Edelist (2014)
Diet	Derbal and Kara (1996), Morales-Nin and Moranta (1997), Lamrini and Bouymajjane (2002) and Lipsky (2012)

22. Alien herbivores (*Siganus luridus* and *S. rivulatus*)

B	Bottom trawl surveys (Edelist et al., 2011; Edelist et al., 2013a)
PB	Z=F+M; M= empirical equation from Pauly (1980). L∞ and K from Bariche (2005)
QB	From the empirical equation of Pauly et al. (1990). W∞ from Abdallah (2002)
Diet	Lundberg and Golani (1995)

23. Earlier alien demersal fishes (*Apogonichthyoides pharaonis, Callionymus filamentosus, Equulites klunzingeri, Lagocephalus spadiceus, L. suezensis, Pelates quadrilineatus, Pempheris rhomboidea, Pomadasys stridens, Sargocentron rubrum, Sillago sihama, Stephanolepis diaspros* and *Torquigener flavimaculosus*)

B	Bottom trawl surveys (Edelist et al., 2011; Edelist et al., 2013a)
PB	Z=F+M; M= empirical equation from Pauly (1980). L∞ and K from Radhakrishnan (1957), Golani and Ben-Tuvia (1985) and Özütok and Avşar (2004)
QB	From the empirical equation of Pauly et al. (1990). W∞ from Taskavak and Bilecenoglu (2001), Cicek et al. (2006), Edelist et al. (2013c) and Edelist (2014)

Diet	Golani et al. (1983), Golani and Diamant (1991), Golani (1993b), Lipsky (2012) and AmtyazSafi et al. (2013)

24. New alien demersal fishes (*Cheilodipterus novemstriatus, Jaydia queketti, J. smithi, Lagocephalus sceleratus, Nemipterus randalli, Ostorhinchus fasciatus, Plotosus lineatus* and *Trypauchen vagina*)

B	Bottom trawl surveys (Edelist et al., 2011; Edelist et al., 2013a)
PB	Z=F+M; M= empirical equation from Pauly (1980). L∞ and K from Pauly (1978), Aydin (2011) and Innal et al. (2015)
QB	From the empirical equation of Pauly et al. (1990). W∞ from Edelist et al. (2013c) and Edelist (2014)
Diet	Golani (2002), Gurlek et al. (2010), Aydin (2011) and Lee (2011)

25. Alien lizardfish (*Saurida lessepsianus*)

B	Bottom trawl surveys (Edelist et al., 2011; Edelist et al., 2013a)
PB	Z=F+M; M= empirical equation from Pauly (1980). L∞ and K from Sanders et al. (1984)
QB	From the empirical equation of Pauly et al. (1990). W∞ from Edelist (2014)
Diet	Golani (1993a) and Lee (2011)

26. Demersal fishes (upper slope) (*Coelorhynchus caelorhinchus, Conger conger, Echiodon dentatus, Helicolenus dactylopterus, Lepidopus caudatus, Lophius budegassa* and *Nettastoma melanurum*)

B	Bottom trawl surveys (Edelist et al., 2011; Edelist et al., 2013a)
PB	Z=F+M; M= empirical equation from Pauly (1980). L∞ and K from Tsimenidis (1980), D'onghia et al. (2000), Nouar (2004) and Filiz et al. (2006)
QB	From the empirical equation of Pauly et al. (1990). W∞ from Tsimenidis and Ondrias (1980), D'onghia et al. (2000), Filiz et al. (2006) and Edelist et al. (2013c).
Diet	Macpherson (1981), Negzaoui-Garali et al. (2008), Sever et al. (2008), Consoli et al. (2010) and Anastasopoulou et al. (2013)

27. Benthopelagic fishes (*Boops boops, Capros aper, Macrorhamphosus scolopax, Spicara maena* and *S. smaris*)

B	Estimated by EE = 0.95
PB	Z=F+M; M= empirical equation from Pauly (1980). L∞ and K from Brethes (1975), Rizkalla (1997), Mehanna (2010) and Josephides (2011)
QB	From the empirical equation of Pauly et al. (1990). W∞ from Gökçe et al. (2009), Mehanna (2010), Josephides (2011) and Edelist et al. (2013c)
Diet	Khoury (1987), Derbal and Kara (2008) and Sever et al. (2013).

28. Mesopelagic fishes (*Chlorophthalmus agassizi, Hygophum hygomii, Lampanyctus crocodilus, L. pusillus, Myctophium punctatum, Stomias boa* and *Vinciguerria poweridae*)

B	Estimated by EE = 0.95

PB	Z=F+M; M= empirical equation from Pauly (1980). L∞ and K from Pauly (1980) and Anastasopoulou et al. (2006)
QB	From the empirical equation of Pauly et al. (1990). W∞ from Merella et al. (1997), Pauly et al. (1998), Morey et al. (2003) and Edelist (2014)
Diet	Scotto di Carlo et al. (1982), Bernal et al. (2013), Sever et al. (2013) and Bernal et al. (2015)

29. Demersal sharks (*Carchinus obscurus, C. plumbeus* and *Mustelus mustelus*)

B	Bottom trawl surveys (Edelist et al., 2011; Edelist et al., 2013a)
PB	Z=F+M; M= empirical equation from Pauly (1980). L∞ and K from Simpfendorfer et al. (2002) and Goosen and Smale (1997)
QB	From the empirical equation of Pauly et al. (1990). W∞ from Filiz and Bilge (2004) and Edelist et al. (2013c)
Diet	Saidi et al. (2007) and Saidi et al. (2009)

30. Rays and skates (*Dasyatis chrysonota, D. pastinaca, Raja miraletus, Rhinobatos rhinobatos, R. cemiculus, Torpedo marmorata* and *T. torpedo*)

B	Bottom trawl surveys (Edelist et al., 2011; Edelist et al., 2013a)
PB	Z=F+M; M= empirical equation from Pauly (1980). L∞ and K from Quignard and Capapé (1974), Abdel-Aziz (1992), Ismen (2003) and Başusta et al. (2008)
QB	From the empirical equation of Pauly et al. (1990). W∞ from Borges et al. (2003), Filiz and Bilge (2004), Başusta et al. (2008), Edelist et al. (2013c) and Edelist (2014)
Diet	Romanelli et al. (2006), Enajjar et al. (2007), Yeldan et al. (2009) and Valls et al. (2011)

31. Small pelagic fishes (*Engraulis encrasicolus, Etrumeus golanii, Sardina pilchardus* and *Sardinella aurita*)

B	Estimated by EE = 0.95
PB	Z=F+M; M= empirical equation from Pauly (1980). L∞ and K from Mater et al. (2003), Yılmaz and Hoşsucu (2003) and STECF (2009)
QB	From the empirical equation of Pauly et al. (1990). W∞ from Voulgaridou and Stergiou (1999), Abdallah (2002) and Edelist et al. (2013c)
Diet	Madkour (2012), Karachle and Stergiou (2013) and Osman et al. (2013)

32. Horse mackerel (*Trachurus mediterraneus, T. picturatus* and *T. trachurus*)

B	Estimated by EE = 0.95
PB	Z=F+M; M= empirical equation from Pauly (1980). L∞ and K from Djabali et al. (1993) and Karlou-Riga (1995)
QB	From the empirical equation of Pauly et al. (1990). W∞ from Djabali et al. (1993) and Ceyhan et al. (2009)
Diet	Bayhan and Sever (2009) and Bayhan et al. (2013)

33. Mackerel (*Scomber colias*)

B	Estimated by EE = 0.95
PB	Z=F+M; M= empirical equation from Pauly (1980). L∞ and K from Bayhan (2007)
QB	From the empirical equation of Pauly et al. (1990). W∞ from Stergiou and Moutopoulos (2001)
Diet	Hattour (2000) and Sever et al. (2006)

34. Native medium pelagic fishes (*Alepes djedaba Caranx crysos, Sphyraena chrysotaenia* and *S. shyraena*)

B	Estimated by EE = 0.90
PB	Z=F+M; M= empirical equation from Pauly (1980). L∞ and K from Goodwin and Johnson (1986), Reuben et al. (1992) and Allam et al. (2004)
QB	From the empirical equation of Pauly et al. (1990). W∞ from Taskavak and Bilecenoglu (2001), Gökçe et al. (2009) and Edelist et al. (2013c)
Diet	Sley et al. (2009) and Kalogirou et al. (2012)

35. Alien medium pelagic fishes (*Decapterus russelli* and *Fistularia commersonii*)

B	Estimated by EE = 0.90
PB	Z=F+M; M= empirical equation from Pauly (1980). L∞ and K from Reuben et al. (1992)
QB	From the empirical equation of Pauly et al. (1990). W∞ from Edelist et al. (2013c) and Edelist (2014)
Diet	Bariche et al. (2009) and Lee (2011)

36. Large pelagic fishes (*Euthynnus alletteratus, Seriola dumerili, Scomberomorus commerson, Xiphias gladius* and *Thunnus thynnus*)

B	ICCAT (2014a) and ICCAT (2014b)
PB	Z=F+M; M= empirical equation from Pauly (1980). L∞ and K from Kedidi et al. (1994), Kahraman and Oray (1998), Aliçli and Oray (2001), Karakulak and Oray (2001) and Kozul et al. (2001)
QB	From the empirical equation of Pauly et al. (1990). W∞ from Kahraman and Oray (1998), Aliçli and Oray (2001), Kozul et al. (2001), Stergiou and Moutopoulos (2001) and Edelist (2014)
Diet	Matallanas et al. (1995), Falautano et al. (2007), Bakhoum (2007), Romeo et al. (2009), Battaglia et al. (2013)

37. Turtles (*Caretta caretta* and *Chelonia mydas*)

B	Estimated by EE = 0.5 based on Levy et al. (2015)
PB	Tsagarakis et al. (2010). Data corrected following Opitz (1996)
QB	Polovina (1984). Data corrected following Opitz (1996)
Diet	Tomas et al. (2001)

38. Sea birds (*Calonectris diomedea, Morus bassanus, Oceanodroma leucoroha, Puffinus griseus, Puffinus yelkouan* and *Stercorarius parasiticus*)

B	Perlman (2013)
PB	Pinnegar (2000)
QB	From the empirical equation of Nilsson and Nilsson (1976). Mean body weight per specie from Del Hoyo et al. (1992)
Diet	Bourgeois et al. (2011) and Neves et al. (2012)

39. Dolphins (*Delphinus delphis, Grampus griseus, Stenella coeruleoalba, Steno bredanensis, Tursiops truncatus* and *Ziphius cavirostris*)

B	Kerem et al. (2014)
PB	Tsagarakis et al. (2010). Data corrected following Opitz (1996)
QB	From the empirical equation of Innes et al. (1987) and Trites et al. (1997). Mean body weight per specie from Carwardine and Camm (1998)
Diet	Kerem et al. (2014) and Astruc (2005)

40. Detritus

B	From the empirical equation of Pauly et al. (1993)

41. Discards

Amount	Cooper et al. (2003), Edelist et al. (2013b), Kerem et al. (2014) and Levy et al. (2015)

Online supplement section 2

Alien species included in this study. Information regarding date of first record is based on Galil (2007), Golani (2010) and other sources (http://www.ciesm.org/online/atlas/). The numbers in the last column of the table identify the functional groups of the Israeli Mediterranean continental shelf model (Table 1 of the manuscript or Supplement Online Information section 1).

Species	Taxonomic group	Habitat	Main prey	Date of first record	Functional group
Alepes djedaba	Fish	Pelagic	fishes	1924	34
Apogonichthyoides pharaonis	Fish	Demersal	-	Before 1947	23
Callionymus filamentosus	Fish	Demersal	benthic invertebrates	before 1953	23
Caranx crysos	Fish	Pelagic	small pelagic fishes	1950's	34
Charybdis longicollis	Crustacean	Demersal	-	1961	11
Cynoglossus sinusarabici	Fish	Demersal	-	Before 1953	18
Decapterus russelli	Fish	Pelagic	small pelagic fishes	2005	35
Equuilites klunzingeri	Fish	Demersal	benthic invertebrates	Before 1934	23
Erugosquilla massavensis	Crustacean	Demersal	-	1938	9
Etrumeus golanii	Fish	Pelagic	zooplankton	1961	31
Fistularia commersoni	Fish	Pelagic	mesopelagic fishes	2000	35
Ixa monodi	Crustacean	Demersal	-	1977	11
Jaydia smithi	Fish	Demersal	benthic invertebrates	2008	24
Jaydia queketti	Fish	Demersal	-	2004	24
Lagocephalus sceleratus	Fish	Demersal	Crustaceans	2004	24
Lagocephalus spadiceus	Fish	Demersal	Crustaceans	1951	23
Lagocephalus suezensis	Fish	Demersal	Crustaceans	1987	23
Marsupenaeus japonicus	Crustacean	Benthopelagic	-	before 1951	9
Metapenaeopsis aegyptica	Crustacean	Benthopelagic	-	1987	9
Metapenaeus monoceros	Crustacean	Benthopelagic	-	before 1951	9
Myra subgranulata	Crustacean	Demersal	-	1929	11
Nemipterus randalli	Fish	Demersal	crustaceans	2005	24
Ostorhinchus fasciatus	Fish	Demersal	benthic invertebrates	recently	24
Pelates quadrilineatus	Fish	Demersal	-	1977	23
Penaeus semisulcatus	Crustacean	Benthopelagic	-	before 1951	9
Pempheris rhomboidea	Fish	Benthic	zooplankton	1979	23
Plotosus lineatus	Fish	Demersal	benthic invertebrates	2001	24
Pomadasys stridens	Fish	Demersal	benthic invertebrates	1971	23
Portunus pelagicus	Crustacean	Demersal	-	before 1924	11
Sargocentron rubrum	Fish	Demersal	benthic invertebrates	1947	23

Saurida lessepsianus	Fish	Demersal	benthic fishes	1952	25
Scomberomorus comerson	Fish	Pelagic	small pelagic fishes	1935	36
Siganus luridus	Fish	Demersal	herbivore	before 1964	22
Siganus rivulatus	Fish	Demersal	herbivore	1924	22
Sillago sihama	Fish	Demersal	benthic invertebrates	1977	23
Sphyraena chrysotaenia	Fish	Pelagic	mesopelagic fishes	before 1931	34
Stephanolepis diaspros	Fish	Demersal	benthic invertebrates	1927	23
Torquigener flavimaculosus	Fish	Demersal	-	1987	23
Trachysalambria curvirostris	Crustacean	Benthopelagic	-	1924-1925	9
Upeneus moluccensis	Fish	Demersal	benthic invertebrates	before 1947	16
Upeneus pori	Fish	Demersal	benthic invertebrates	before 1953	16

Online supplement section 3

Outputs of the models to consider that the model is balanced. R/A = respiration/assimilation ratio; R/B = respiration/biomass ratio (year^{-1}); and NE = net efficiency. In addition, EE and P/Q were checked (Table 1 of the manuscript). Dem. = demersal

Functional group	R/A		R/B		P/R		NE	
	1990s	2010s	1990s	2010s	1990s	2010s	1990s	2010s
Phytoplankton	-	-	-	-	-	-	-	-
Benthic primary producers	-	-	-	-	-	-	-	-
Micro and Mesozooplankton	0.45	0.45	18.67	18.97	1.22	1.22	0.55	0.55
Macrozooplankton	0.50	0.50	17.15	17.43	1.00	1.00	0.50	0.50
Gelatinous plankton	0.66	0.66	29.76	30.24	0.52	0.52	0.34	0.34
Polychaetes	0.61	0.61	8.40	8.53	0.63	0.63	0.39	0.39
Suprabenthos	0.70	0.70	28.15	28.24	0.42	0.42	0.30	0.30
Native shrimps	0.60	0.60	4.71	4.73	0.66	0.66	0.40	0.40
Alien shrimps	0.62	0.62	4.99	5.01	0.62	0.62	0.38	0.38
Native crabs	0.62	0.62	4.60	4.63	0.61	0.61	0.38	0.38
Alien crabs	0.63	0.63	4.84	4.86	0.58	0.58	0.37	0.37
Benthic invertebrates	0.62	0.63	5.45	5.47	0.60	0.60	0.38	0.38
Benthic cephalopods	0.68	0.68	4.02	4.03	0.47	0.47	0.32	0.32
Benthopelagic cephalopods	0.88	0.88	18.68	18.74	0.13	0.13	0.12	0.12
Mullets	0.72	0.72	4.80	4.83	0.39	0.39	0.28	0.28
Goatfishes	0.72	0.73	4.95	4.99	0.38	0.38	0.27	0.28
Hake	0.81	0.81	3.75	3.78	0.24	0.24	0.19	0.19
Flatfishes	0.77	0.77	5.06	5.08	0.30	0.30	0.23	0.23
Rocky fishes	0.70	0.70	4.01	4.03	0.43	0.43	0.30	0.30
Small native dem. fishes	0.72	0.72	3.81	3.84	0.39	0.39	0.28	0.28
Large native dem. fishes	0.77	0.77	3.76	3.78	0.29	0.29	0.23	0.23
Alien herbivores	0.69	0.69	3.79	3.81	0.45	0.45	0.31	0.31
Earlier alien dem. fishes	0.71	0.71	3.85	3.86	0.41	0.41	0.29	0.29
New alien dem. fishes	0.72	0.72	3.85	3.85	0.39	0.39	0.28	0.28
Alien lizardfish	0.79	0.79	4.43	4.45	0.27	0.27	0.21	0.21
Demersal fishes (upper slope)	0.79	0.79	3.73	3.74	0.27	0.27	0.21	0.21
Benthopelagic fishes	0.75	0.75	6.08	6.17	0.33	0.33	0.25	0.25
Mesopelagic fishes	0.75	0.75	5.22	5.24	0.33	0.33	0.25	0.25
Demersal sharks	0.81	0.81	3.03	3.04	0.23	0.23	0.19	0.19
Rays and skates	0.79	0.79	3.62	3.63	0.26	0.26	0.21	0.21
Small pelagic fishes	0.75	0.75	7.19	7.36	0.33	0.33	0.25	0.25
Horse mackerel	0.78	0.78	4.97	5.04	0.29	0.29	0.23	0.23
Mackerel	0.78	0.78	4.89	4.96	0.29	0.29	0.23	0.23
Native medium pelagic fishes	0.85	0.85	4.80	4.81	0.18	0.18	0.15	0.15
Alien medium pelagic fishes	0.85	0.85	4.83	4.83	0.18	0.18	0.15	0.15
Large pelagic fishes	0.79	0.79	3.38	3.40	0.27	0.27	0.21	0.21
Turtles	0.92	0.92	2.06	2.09	0.08	0.08	0.08	0.08

Sea birds	0.92	0.92	55.08	55.08	0.09	0.09	0.08	0.08
Dolphins	0.99	0.99	8.67	8.67	0.01	0.01	0.01	0.01
Detritus	-	-	-	-	-	-	-	-
Discards	-	-	-	-	-	-	-	-

Online supplement section 4

The pre-balancing analysis (PREBAL) was designed to assess the data quality before the mass balance process and/or dynamic simulations are developed with Ecopath and Ecosim food web models in order to ensure that the input data are reasonable and robust and to indirectly verify model accuracy and robustness (Link, 2010). Therefore, once the input data has been collected and analysed, it is needed to test the model through the PREBAL analysis (Heymans et al., 2016).

Link (2010) describes a set of diagnostics that can be tested thought graphical representation, including biomass and vital rates vital rates across taxa and trophic levels, biomass and vital rates ratios, and total production and removals.

In this study, we checked the diagnostics related to biomass and vital rates (P/B and Q/B) across taxa and trophic levels. Biomass and vital rates of each functional group were plotted along the x-axis in order of decreasing trophic level. In the biomass diagnostic, we check the two rules that Link (2010) proposed: 1) the range of biomass should span 5-7 orders of magnitude and 2) the slope should decline by 5-10% with increasing trophic level across all the taxa. In addition, biomass and vital rates of predators should be lower than prey. For all of these diagnostics, groups with values that greatly exceed or fall below the general trend were examined.

Results of the PREBAL analysis in both time periods are shown in Fig. 1 (1990s) and Fig. 2 (2010s). Both models complied with the two rules that we checked for biomass as the range of biomass spanned 5 orders of magnitude and the slope declined by 5.5% and 6.1% in 1990s and 2010s time periods, respectively.

The overall diagnostics applied to our models showed general coherent reduction of parameters values with increasing trophic levels (Fig. 1 and 2). Regarding biomass, in 2010s period all the functional groups were represented, while in the 1990s period, new alien demersal fishes (functional group or F.G. 24) and alien medium pelagic fishes (F.G. 35) were not present because they did not invaded the ecosystem yet (Fig. 1). In both models, sea birds (F.G. 38) presented low biomass despite their trophic position (Fig. 1a and 2a). This is due to their uncertainty of biomass estimates and their trophic positions in Ecopath models, as an important part of their diet are discards, which EwE considers as a detritus group with TL = 1.

The results also showed that phytoplankton (F.G.1), micro and mesozooplankton (F.G. 3), polychaetes (F.G. 6), benthic invertebrates (F.G. 12), small pelagic fishes (F.G. 31), small native demersal fishes (F.G. 20), native medium pelagic fishes (F.G. 34) and large pelagic fishes (F.G. 36) could potentially be overestimated, while benthic primary producers (F.G. 2), alien herbivorous (F.G. 22), demersal fishes in the upper slope (F.G. 26) and dolphins could be underestimated in both time periods (Fig. 1a and 2a). As expected by Link (2010), groups at the two extremes (in the ICS models, phytoplankton (F.G. 1) and dolphins (F.G. 39)) tend to present out of trend biomass values. In the case of small native demersal fishes (F.G. 20), we can justify their higher biomass due to the large number of species included in this group. For native medium pelagic fishes (F.G. 34) and large pelagic fishes (F.G. 36), the last one with biomass estimate based on stock assessment, we could expect lower biomass but the requirements of the model imposed higher biomass mainly due to their large catch, as their predation mortalities showed low values (Table 2 in the manuscript). For the remainder groups, biomass estimates were calculated by the model, showing the need of stock assessment for these groups. In absence of biomass estimates, adjustments on the diet matrix were needed in order to ensure the correct diagnostic in PREBAL.

Moreover, in the 1990s period, large native demersal fishes (F.G. 21) presented a high biomass. We could assume that their biomass estimate was overestimated but, as in other cases, the large catch imposed high requirements to the model, as their predation mortalities showed low values (Table 1 in the manuscript). This group did not show a high biomass in the 2010s period due to its overexploitation during the period.

Benthic primary producers (F.G. 2) and alien herbivorous (F.G. 22) presented low biomass values as these groups live exclusively in the shallow waters, while the model included depths from 0 to 200 m. Therefore, we could expect low biomasses for these groups. Demersal fishes in the upper slope (F.G. 26) also showed low biomass estimates. These could be related to the lower number of hauls in these strata on the trawl survey used to estimate biomasses and/or to the deep-sea faunal and biomass impoverishment of the Eastern Mediterranean Sea compared to the western basin (Goren and Galil, 2002).

In addition, alien crabs (F.G. 11) and native crabs (F.G. 10) showed low biomass in 1990s and 2010s time periods, respectively (Fig. 1a and 2a). We could expect these

results as the biomass of alien crabs (F.G. 11) increased during the periods while the biomass of native crabs decreased although the native crabs (F.G. 10) did not present high values in 1990s period.

Finally, mullets (F.G. 15) and hake (F.G. 17) showed low biomass estimates in the 2010s period as a result of overexploitation and niche displacement to deeper water by their alien competitors (goatfishes (F.G. 16) and alien lizardfish (F.G. 25)) (Galil and Zenetos, 2002; Rilov and Galil, 2009).

In the ICS model, P/B ratios for alien herbivorous (F.G. 22), benthic invertebrates (F.G. 12), turtles (F.G. 37) and dolphins (F.G. 39) were low, while those for phytoplankton (F.G. 1), micro and mesozooplankton (F.G. 3), macrozooplankton (F.G. 4), gelatinous plankton (F.G. 5) and benthopelagic cephalopods (F.G. 14) were high (Fig. 1b and 2b). Initial P/B values for macrozooplankton (F.G. 4), polychaetes (F.G. 6) and suprabenthos (F.G. 7) were modified as we were able to find better parameters estimates taking into account the PREBAL analysis. For macrozooplankton (F.G. 4), initial P/B value was 22.92 (from Labat and Cuzin-Roudy (1996)), and as this value was high we decided to apply a P/Q of 0.3 taking into account the P/Q of 0.33 for micro and mesozooplankton (F.G. 3). For polychaetes (F.G. 6) and suprabenthos (F.G. 7), initial P/B values were weighted values of different species but they were low. Therefore, we decided to use the maximum value. For the other problematic groups, in absence of better parameter estimates, we decided to use common values in other Ecopath models set up in the Mediterranean Sea (Coll et al., 2006; Coll et al., 2007; Tsagarakis et al., 2010; Corrales et al., 2015).

Finally, in the ICS model, Q/B ratios for alien herbivorous (F.G. 22), benthic invertebrates (F.G. 12), shrimps (F.G. 8 and 9), crabs (F.G. 10 and 11), rocky fishes (F.G. 19) and turtles (F.G. 37) were low, while those for micro and mesozooplankton (F.G. 3), macrozooplankton (F.G. 4), gelatinous plankton (F.G. 5), suprabenthos (F.G. 7), benthopelagic cephalopods (F.G. 14), sea birds (F.G. 38) and dolphins were high. Q/B ratios for shrimps (F.G. 8 and 9), crabs (F.G. 10 and 11), rocky fishes (F.G. 19), seabirds (F.G. 38) and dolphins (F.G. 39) were maintained as they came from empirical equations (See Supplement Online Information section 1). Initial Q/B ratios for alien herbivorous (F.G. 22) and demersal sharks (F.G. 29) were modified as those had low and high P/Q ratios respectively. For alien herbivorous (F.G. 22) we decided to apply a

high P/Q ratio (0.25) and for demersal sharks (F.G. 29) a low P/Q (0.15), taking into account their trophic position and ecological principles. The other problematic groups, in absence of better parameter estimates and as for P/B ratios, we decided to use common values in other Ecopath models set up in the Mediterranean Sea (Coll et al., 2006; Coll et al., 2007; Tsagarakis et al., 2010; Corrales et al., 2015).

To conclude, the PREBAL analysis applied to the ICS model showed that in general the models complied general ecological principles. Moreover, it showed that for some groups, better information is required, mainly for planktonic groups, which should inform future scientific research objectives in the area. As new information is being generated, the models could be updated and therefore, its quality and reliability increased. In addition, with the calibration of the 1990s model to available historical data, some of these parameters should be rechecked and therefore data input of the model may improve.

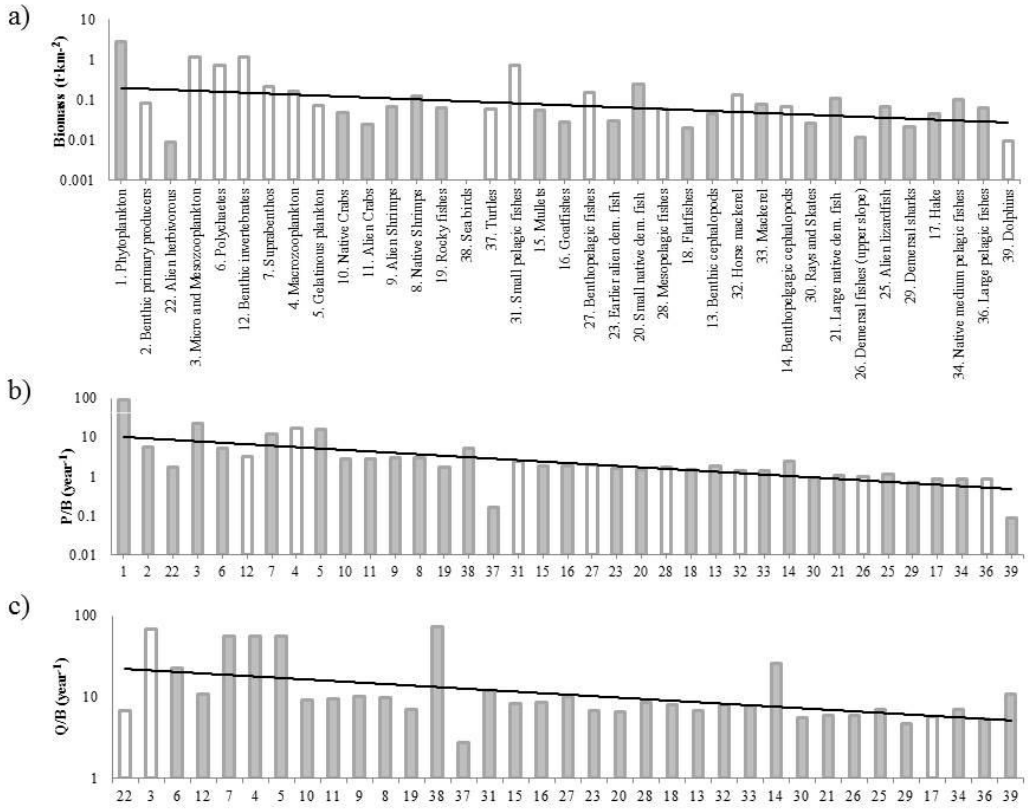


Fig. 1. PREBAL of the Israeli Mediterranean coast in the 1990s period plotting Biomass estimates ($t \cdot km^{-2}$) (a), Production/Biomass ratio ($year^{-1}$) (b), and Consumption/Biomass ratio ($year^{-1}$) (c) on a log scale vs species ranked by trophic level, from the lowest to highest trophic level. Trend

lines are given. Solid bars indicate input to the model while white bars indicate model estimations or changed values during the PREBAL analysis. Numbers in (b) and (c) identify the functional groups of the model (a).

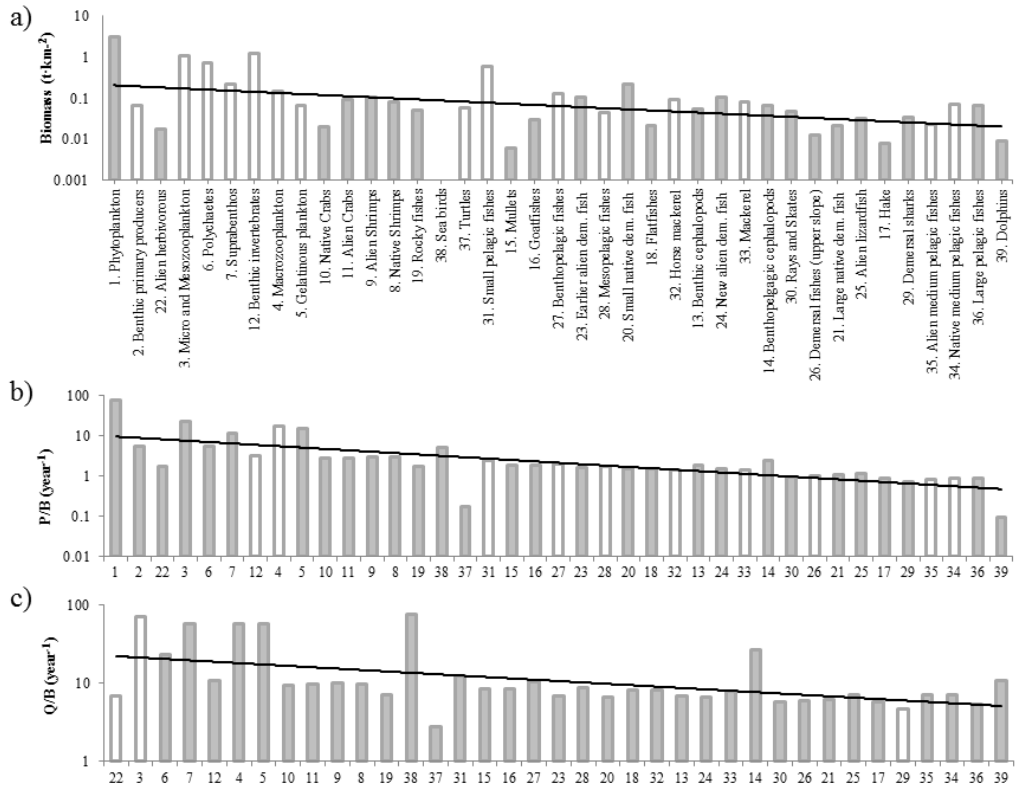


Fig. 2. PREBAL of the Israeli Mediterranean coast in the 2010s period plotting Biomass estimates ($t \cdot km^{-2}$) (a), Production/Biomass ratio ($year^{-1}$) (b), and Consumption/Biomass ratio ($year^{-1}$) (c) on a log scale vs species ranked by trophic level, from the lowest to highest trophic level. Trend lines are given. Solid bars indicate input to the model while white bars indicate model estimations or changed values during the PREBAL analysis. Numbers in (b) and (c) identify the functional groups of the model (a).

Online supplement section 5

Table 1. Diet composition matrix for the ICS model in the 1990-1994 period. Cells in grey are values lower than 0.01. The numbers identify the functional groups of the model (Table 1 of the manuscript or Supplement Online Information section 1).

	3	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**	**13**	**14**	**15**	**16**	**17**	**18**
1	0.70	0.20	0.10							0.08						
2										0.01						
3	0.05	0.65	0.74		0.08	0.05	0.05			0.08		0.06				
4		0.05	0.05			0.15	0.15	0.12	0.12			0.10			0.02	0.10
5			0.01													
6				0.05	0.02	0.23	0.23	0.28	0.28		0.16	0.14	0.32	0.26	0.02	0.21
7					0.03	0.16	0.16	0.10	0.10	0.01	0.19	0.13	0.16	0.19	0.05	0.26
8								0.01	0.01		0.06	0.02	0.08	0.05	0.04	0.02
9											0.03	0.01	0.04	0.08	0.02	0.02
10											0.02		0.03	0.03	0.01	0.02
11											0.01		0.01	0.02	0.01	0.01
12					0.02	0.19	0.19	0.21	0.21	0.05	0.47	0.17	0.24	0.25	0.07	0.34
13											0.01				0.04	
14												0.01			0.05	
15															0.02	
16																
17															0.02	
18																
19											0.02					0.02
20											0.02				0.08	0.01
21															0.02	
22																
23																
25															0.01	
26															0.01	
27												0.02			0.03	
28												0.01			0.02	

29																			
30																			
31												0.32			0.35				
32															0.03				
33															0.02				
34															0.04				
36																			
37																			
38																			
39																			
40	0.25	0.10	0.10	0.95	0.85	0.22	0.22	0.28	0.28	0.77			0.13	0.13	0.02				
41																			
Import																			
Total	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1			

	19	**20**	**21**	**22**	**23**	**25**	**26**	**27**	**28**	**29**	**30**	**31**	**32**	**33**	**34**	**36**	**37**	**38**	**39**
1				0.30				0.02				0.02							
2	0.07	0.02		0.60															
3	0.05	0.02			0.03			0.78	0.61			0.91	0.63	0.37					
4	0.05	0.02			0.03		0.05	0.11	0.25	0.04	0.04	0.06	0.27	0.46	0.04	0.05			
5		0.01					0.03	0.04	0.05			0.01	0.02	0.06	0.02	0.03	0.09		
6	0.21	0.22	0.09		0.25	0.04	0.11	0.05	0.09	0.03	0.11		0.04	0.02	0.02				
7	0.21	0.24	0.17		0.24	0.13	0.19			0.07	0.14								
8		0.04	0.09		0.06	0.05	0.08			0.03	0.06		0.01			0.02			
9		0.03	0.04		0.03	0.03					0.01					0.01			
10	0.01	0.02	0.02		0.02	0.03	0.05			0.03	0.06						0.05		
11					0.01	0.01				0.02	0.02						0.01		
12	0.31	0.29	0.17		0.29	0.18	0.16			0.10	0.28						0.30		
13		0.01	0.02		0.01	0.01	0.01			0.05	0.01								
14			0.02			0.01	0.01			0.03	0.01				0.03	0.12		0.01	0.18
15			0.02			0.02	0.01			0.04									
16			0.01			0.01				0.02									
17							0.03			0.03						0.01			0.06

18							0.02			0.02	0.01								
19		0.03	0.02			0.02					0.02								
20		0.01	0.09			0.05	0.02			0.07	0.04				0.04	0.06			0.03
21			0.02			0.01	0.01			0.04						0.01			0.01
22			0.01																
23			0.01			0.01				0.02	0.01								
25			0.02							0.02						0.01			0.01
26							0.02			0.02									0.02
27			0.05			0.03	0.03			0.04	0.04				0.18	0.04			0.04
28							0.06			0.04	0.02				0.07	0.02			0.02
29																			
30							0.02			0.02									
31		0.02	0.13			0.27	0.10			0.18	0.09		0.02	0.09	0.44	0.32		0.29	0.40
32						0.03				0.03	0.02				0.08	0.08			0.08
33						0.01				0.02					0.06	0.06			0.06
34						0.03				0.02	0.01					0.03			
36																0.01			
37																			
38																			
39																			
40	0.07	0.02		0.10	0.01														
41						0.01											0.15	0.30	
Import																0.10	0.40	0.40	0.10
Total	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1

Table 2. Diet composition matrix for the ICS model in the 2008-2012 period. Cells in grey are values lower than 0.01. The numbers identify the functional groups of the model (Table 1 of the manuscript or Supplement Online Information section 1).

	3	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**	**13**	**14**	**15**	**16**	**17**	**18**	**19**	**20**	**21**
1	0.70	0.20	0.10							0.08									
2										0.01							0.07	0.02	
3	0.05	0.65	0.74		0.08	0.05	0.05			0.09		0.08					0.06	0.03	
4		0.05	0.05			0.15	0.15	0.12	0.13			0.10			0.02	0.10	0.06	0.03	
5			0.01															0.01	
6				0.05	0.02	0.23	0.23	0.28	0.27		0.17	0.13	0.31	0.25	0.02	0.21	0.20	0.20	0.07
7					0.03	0.16	0.16	0.10	0.10	0.01	0.18	0.12	0.16	0.18	0.06	0.26	0.20	0.22	0.13
8								0.01	0.01		0.05	0.01	0.07	0.04	0.04	0.02		0.03	0.05
9											0.05	0.02	0.06	0.10	0.02	0.02		0.04	0.07
10													0.02		0.01				
11											0.03		0.02	0.06	0.01	0.02	0.02	0.03	0.03
12					0.02	0.19	0.19	0.21	0.21	0.05	0.44	0.17	0.24	0.25	0.07	0.33	0.31	0.27	0.15
13											0.01				0.04			0.02	0.03
14												0.01			0.06				0.03
15															0.02				0.01
16																			0.02
17															0.01				
18															0.01				0.01
19											0.01					0.01		0.02	0.02
20											0.02				0.07	0.01		0.01	0.08
21															0.01				
22																			0.01
23											0.02				0.02				0.06
24											0.01				0.02				0.06
25															0.01				0.01
26															0.02				
27												0.02			0.03				0.04
28												0.01			0.03				
29																			

30																			
31												0.32			0.32			0.02	0.12
32															0.02				
33															0.02				
34															0.03				
35																			
36																			
37																			
38																			
39																			
40	0.25	0.10	0.10	0.95	0.85	0.22	0.22	0.28	0.28	0.76			0.13	0.13	0.01		0.07	0.02	
41																			
Import																			
Total	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1

	22	**23**	**24**	**25**	**26**	**27**	**28**	**29**	**30**	**31**	**32**	**33**	**34**	**35**	**36**	**37**	**38**	**39**
1	0.30					0.02				0.02								
2	0.60																	
3		0.03	0.01			0.78	0.61			0.91	0.63	0.37						
4		0.03	0.01		0.05	0.11	0.25	0.04	0.04	0.06	0.27	0.46	0.04	0.05	0.05			
5					0.03	0.04	0.05			0.01	0.04	0.07	0.02	0.02	0.03	0.09		
6		0.24	0.20	0.03	0.12	0.05	0.09	0.02	0.11		0.02	0.02	0.02	0.02				
7		0.24	0.26	0.10	0.19			0.05	0.13									
8		0.03	0.02	0.03	0.08			0.02	0.05						0.02			
9		0.07	0.05	0.04				0.02	0.04					0.01	0.02			
10		0.01			0.05			0.02	0.02							0.03		
11		0.02	0.05	0.03				0.04	0.05							0.04		
12		0.29	0.25	0.15	0.15			0.09	0.26					0.01		0.30		
13		0.01	0.02	0.02	0.01			0.05	0.01									
14				0.02	0.02			0.04	0.01				0.04	0.04	0.14		0.01	0.18
15																		
16				0.03				0.03										
17					0.01													0.01

18				0.01	0.02			0.03	0.01										
19			0.02	0.02					0.02										
20			0.01	0.05	0.03			0.06	0.04				0.04	0.04	0.06				0.02
21								0.01											
22				0.01															
23			0.01	0.07				0.06	0.03				0.03	0.03	0.03				0.04
24				0.07				0.06	0.03				0.02	0.02	0.03				0.04
25								0.02											0.01
26					0.03			0.02											0.02
27				0.04	0.03			0.04	0.04				0.16	0.17	0.03				0.04
28					0.06			0.04	0.02				0.06	0.01	0.02				0.02
29					0.01														
30					0.02			0.02											
31			0.07	0.22	0.10			0.16	0.07		0.02	0.09	0.42	0.44	0.30		0.29	0.38	
32				0.02				0.02	0.01				0.06	0.06	0.06				0.07
33				0.02				0.02					0.06	0.06	0.07				0.07
34				0.01				0.01							0.02				
35																			
36																			
37																			
38																			
39																			
40	0.10	0.01	0.01																
41				0.01												0.15	0.30		
Import															0.10	0.40	0.40	0.10	
Total	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	

Online supplement section 6

Confidence intervals and index values used to describe the uncertainty for each input parameter of the balanced Ecopath model.

Parameter	Option	Index value	CI (%)
Biomass			
	Estimated by Ecopath	0.0	±80
	From other model	0.0	±80
	Guesstimate	0.0	±80
	Approximate or indirect method	0.4	±50
	Sampling/locally, low precision	0.6	±40
	Sampling/locally, high precision	0.8	±10
PB/ and Q/B			
	Estimated by Ecopath	0.0	±80
	Guesstimated	0.1	±70
	From other model	0.2	±60
	Empirical relationship	0.5	±50
	Similar species, similar system, low precision	0.6	±40
	Similar species, same system, low precision	0.7	±30
	Same species, similar system, low precision	0.8	±20
	Same species, same system, high precision	1	±10
Diet			
	From other model	0.1	±80
	General knowledge of related groups/species	0.1	±70
	General knowledge for same groups/species	0.2	±60
	Qualitative diet composition study	0.5	±50
	Quantitative (but limited) diet composition study	0.6	±40
	Quantitative (detailed) diet composition study	0.7	±30
Catch			
	From other model	0.1	±90
	National statistics	0.5	±50
	Intermediate (local study and national statistics)	0.6	±40
	Local study, low precision/incomplete data	0.7	±30
	Local study, high precision/complete data	0.9	±10

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